

Illusory Traits as the Number of Timepoints Increases: a New Method to Account for Unobserved Time-Varying Confounding

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Abstract

The cross-lagged panel model (CLPM) is a popular option for researchers in the social sciences interested in multivariate longitudinal data. Often criticized as being too restrictive in assumptions concerning confounding, this paper presents an extension of the CLPM aimed at controlling for both time-varying and time-invariant confounding. This development is heavily inspired by the ‘illusory trait’ finding, showing that the RI-CLPM can not only correct for the time-invariant confounding it was designed to correct for, but also for time-varying confounding. The performance of this new model, the segmented random intercept cross-lagged panel model (SRI-CLPM) is examined in a series of Monte Carlo simulations. These show the model outperforming the RI-CLPM and CLPM in retrieving less biased estimates of cross-lagged parameters, at a slight increase in standard errors. This effect becomes more pronounced as more timepoints are considered, as the RI-CLPM starts to be unable to correct for time-varying confounding. A possible further extension of the SRI-CLPM is also explored, aimed at separating the variance associated with time-varying and time-invariant confounding.

Keywords: CLPM, RI-CLPM, Cross-Lagged Panel Modeling, Illusory Traits, SRI-CLPM, Time-Varying Confounding

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When researchers are concerned with studying how two variables interact over time, a helpful technique is the estimation of cross-lagged parameters ([Orth, Clark, Donnellan, & Robins, 2021](#)). Cross-lagged parameters predict the change in an outcome variable based on a previous measurement of the other variable, while controlling for the score on the outcome variable at the preceding occasion through an autoregressive parameter. Models that include these parameters, such as the cross-lagged panel model (CLPM, [Duncan, 1969](#)), are widely used in the social sciences. A major concern when using these estimates for causal inference purposes is the possibility of them being affected by unobserved confounding ([Hamaker, Kuiper, & Grasman, 2015](#); [Rogosa, 1980](#)).

Confounding refers to the process by which the relationship between two variables is at least partially explained through the causal effects a third variable has on them both. In longitudinal research, we can broadly classify confounding into two types: time-varying and time-invariant. The CLPM has been criticized as having its cross-lagged parameter estimation confounded by stable, trait-like between-person differences ([Hamaker et al., 2015](#); [Lucas, 2023](#)). Models have therefore been developed that aim to control for these time-invariant confounders, such as the random-intercept cross-lagged panel model (RI-CLPM, [Hamaker et al., 2015](#)) and alternatives such as the Trait-State-Error model ([Kenny & Zautra, 1995](#)). These models are designed to take away the influence of time-invariant confounding by modeling them through the inclusion of latent variables. In the RI-CLPM this is achieved by specifying a random intercept, that is, a latent variable with factor loadings constrained to 1 on all instances of a variable. This essentially centers the variables around the mean value of an individual case (assuming that mean does not change over time), a technique inspired by the multilevel modeling literature. In doing so the model separates stable between-person variation (e.g. genetic predisposition, personality traits) from within-person fluctuations. This is then argued to be a better representation of the theoretical model, compared to the simpler CLPM which assumes individual cases have the same

underlying mean ([Lucas, 2023](#)). These models, and other related approaches, have been widely adopted by researchers. An understudied topic however, is how to control for unobserved time-varying confounding.

One approach in dealing with time-varying confounding, developed by Kenny and McCoach ([2025](#)), is the Latent Time-Varying Confounder model (LTVC). This is a model that aims to test the alternative hypothesis that all covariation between two variables can be explained by a time-varying confounder. The authors claim that a well-fitting LTVC only means that the possibility of all covariation being explained by a time-varying confounder cannot be ruled out ([Kenny & McCoach, 2025](#)). While useful, this means it is not able to retrieve parameter estimates for the model of theoretical interest while controlling for time-varying confounding. The authors do show a model, the LTVC-CLPM, which as the name suggests is a combination of both models, that aims to remedy this limitation. They note however, that they were not able to validate the model due to poor model fit and anomalous parameter estimates. In this paper, we will take a different approach to modeling time-varying confounding by considering systems of time-varying confounding, rather than one underlying confounding variable.

A recent paper by Bailey et al. ([2024](#)) investigated the performance of a bivariate RI-CLPM in the presence of systems of time-varying confounding. They demonstrated that finding a significant random intercept variance does not necessarily imply there was a time-invariant confounder in the data generating process. Instead, when there are many different time-varying confounders present, each having only a small effect on each other and the variables of interest, fitting a RI-CLPM without including the confounders estimates a significant random intercept variance. Therefore the model seems to find evidence for the existence of a stable, between person 'trait'. Bailey et al. ([2024](#)) refer to this phenomenon as 'illusory traits', underlining these are not truly underlying between person differences. Surprisingly, the addition of the random intercept in the RI-CLPM controls for the presence of this time-varying confounding, allowing the model to retrieve less biased estimates of cross-lagged parameters than the CLPM. Therefore, Bailey and colleagues caution researchers against interpreting significant random intercept variance as

evidence of a stable trait, yet still advise researchers to choose a RI-CLPM over a CLPM. The question arising is what happens to these illusory traits in the long run. Bailey and colleagues (2024) simulated datasets with only five timepoints. The authors show the process behind estimating the illusory traits by plotting the rank order stability with and without the time-varying confounding. The autocorrelation between measurements decreases asymptotically to zero, but this decay is slowed by the time-varying confounding. They do not show however, how the performance of the RI-CLPM develops as more timepoints are considered. Since the data were generated according to a stationary process, meaning the underlying distributions of the variables do not change over time (Kirchgässner & Wolters, 2008, pp. 12–12), we expect any excess stability introduced by the time-varying confounders to dissipate over time. In other words, given enough time, the process will return to fluctuate around its mean value.

Investigating what happens when the number of timepoints increases is especially relevant given the recent developments which allow for the collection of more and more measurements through devices like smartphones and smartwatches. This type of data is referred to as intensive longitudinal data (ILD) and is characterized by many measurements of variables densely packed in time (Hamaker, Asparouhov, & Muthén, 2023). We can contrast this with panel data, which are characterized by only a few measurements in study lengths measured in weeks or months, rather than days. This density of measurements allows researchers to study dynamics on small timespans, such as within a single day (Hamaker, 2025). However, this can be perceived as burdensome to participants, given the frequency with which a participant is asked to answer a psychological measure. It is therefore also doubtful whether this type of data collection allows for the measurement of all potential confounders of importance on every timepoint, due to the substantial extra burden this poses to participants. Given the interest in these studies and their increased feasibility, assessing the performance of commonly used models with confounding in ILD is of substantial importance. The aim of this paper is twofold. First, we aim to investigate the expectation that the RI-CLPM is a less effective way of dealing with time-varying unobserved confounders as the number of timepoints increases, and that in the long-run its performance

approaches that of the CLPM. Second, we aim to design a model that can control for time-varying confounding by introducing a model we refer to as the Segmented Random Intercept Cross-Lagged Panel Model (SRI-CLPM). This model aims to control for time-varying confounding by specifying random intercepts for segments of the total timespan of the study, inspired by the illusory trait finding. The rest of this paper is structured as follows. We start by extending the simulation done by Bailey et al. (2024) to scenarios with more timepoints to see whether the illusory traits indeed largely disappear in the long run; we expect that this will go hand-in-hand with an increase in bias in the cross-lagged parameters which will approximate that of the CLPM as the number of timepoints increases. Subsequently, we present the SRI-CLPM as a way to leverage the finding of Bailey et al. in the context of increasing number of timepoints, and perform an explorative study of an extension intended to be able to separate time-invariant and time-varying confounding. We simulate data and assess the performance of the SRI-CLPM under varying segment timespans and covariance constraints. Lastly, we provide an overall discussion and look ahead to possible future developments.

The aim of this paper is threefold. First, we will investigate whether our expectation is correct that the RI-CLPM becomes a less effective way of dealing with time-varying unobserved confounders as the number of timepoints increases, and that in the long-run its performance approaches that of the CLPM. Second, we will present a new model to handle time-varying confounding as the number of timepoints increases: the Segmented Random Intercept Cross-Lagged Panel Model (SRI-CLPM), which controls for time-varying confounding through the inclusion of random intercepts for segments of the total timespan of the study. Third, we will present an extension of the SRI-CLPM with a global random intercept (SGRI-CLPM), aimed at separating time-invariant from time-varying confounding. The remainder of this paper is structured as follows. We start by extending the simulation done by Bailey et al. (2024) to scenarios with more timepoints to see whether the illusory traits indeed largely disappear in the long run; we expect that this will go hand-in-hand with an increase in bias in the cross-lagged parameters which will approximate that of the CLPM as the number of timepoints increases.

Subsequently, we present the SRI-CLPM as a way to leverage the finding of Bailey et al. in the context of an increasing number of timepoints, and assess its ability to retrieve unbiased estimates of the cross-lagged parameters under time-varying confounding. Thereafter, we present the SGRI-CLPM and investigate whether it is possible to separate confounding into a time-varying and time-invariant component using this extension. Lastly, we provide an overall discussion and look ahead to possible future developments.

Illusory Traits in the RI-CLPM as the Number of Timepoints Increases

We hypothesize that as the number of timepoints increases, the illusory traits will disappear (that is, their variance will go to zero), and that this development goes hand-in-hand with a deterioration of the ability of the RI-CLPM to retrieve relatively unbiased cross-lagged parameter estimates. We therefore build on the simulations done by Bailey et al. (2024) by considering the effect of more timepoints. In addition, we also vary the number of time-varying confounders and the strength of the autoregressive effects across all variables to illustrate the role of temporal stability in the observed processes.

Extending the Simulation by Bailey and colleagues

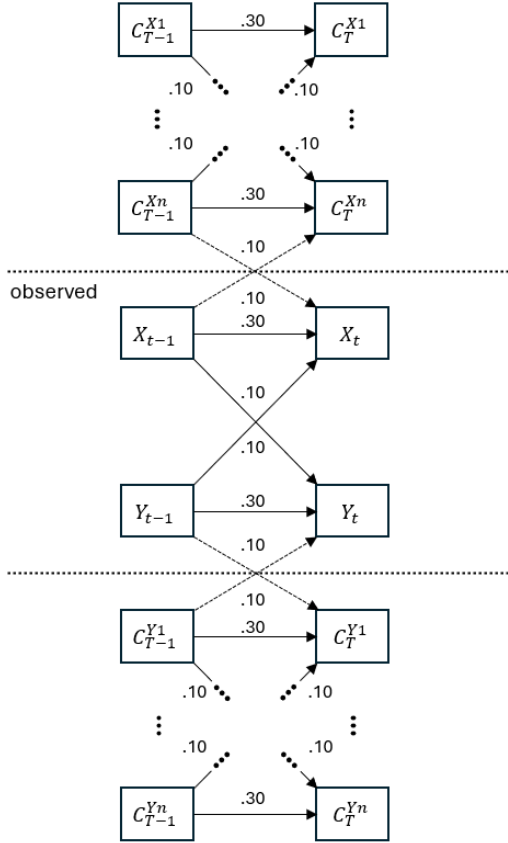
The data generating mechanism is adapted from the original illusory trait paper by Bailey et al. (2024), largely following their provided R-code. We use the same underlying parameter values they specified as the defaults, only deviating from this to investigate particular properties. We introduce their data generating mechanism here, but refer to the original paper for a more in-depth discussion of the parameter value choices. A schematic representation of the data generating mechanism can be found in Figure 1. The data generating mechanism consists of a bivariate CLPM with the addition of systems of time-varying unobserved confounding. The relationships between the observed variables X and Y are expressed in the autoregressive effects and the cross-lagged effects.

In the first simulation, we generate data according to the data generating mechanism, with conditions varying the number of unobserved confounders on both variables and varying the autoregressive effects for all variables (observed or unobserved). We will then attempt to

accurately retrieve estimates of the cross-lagged parameters under confounding, fitting models only on the observed variables X and Y . We vary the number of unobserved confounders on each variable to be either 4 or 6, resulting in a total number of 8 or 12 respectively. The system (or grouping) of unobserved confounders on one variable has the same parameter values with themselves and their observed variable as before; cross-lagged effects of .1 and autoregressive effects varying to be .2 or .3. Additionally, the two systems weakly influence each other, with all confounders on X having a lagged effect of .01 on the confounders on Y and vice versa. Time-specific residual variances were specified to be .5. After specifying the model, we sample from the stationary covariance matrix of a single timepoint (see equation 2.1.39 in [Lütkepohl, 2007](#)) following the procedure in the original paper. This approach has the benefit of not requiring to start up the process and wait for it to reach its stationary state. We take a large sample of 20 thousand cases, ensuring we can clearly estimate the degree of bias in the cross-lagged parameters without the influence of sampling variation.

Figure 1

Schematic representation of the data generating mechanism adapted from Bailey et al. Triple dots represent left out intermediary variables for visual clarity.



Stationarity & Ergodicity

As mentioned before, we simulate data from a stationary process, meaning the properties of the underlying data distributions do not change over time. Stationarity is often defined in terms of the first two moments, which implies that the mean, variances and covariances of a process are invariant over time; in this case, the process is said to be weakly stationary ([Kirchgässner & Wolters, 2008, pp. 22–13](#)). In the current context, the data that are generated also conform to strict stationarity, meaning the third and fourth moments also do not change over time. This is automatically the case when generating data along a multivariate normal distribution, since its third central moment is zero (the distribution is symmetric), and its fourth central moment is a function of the covariances, which do not change over time as follows from weak stationarity. To

ensure the data generating process is stationary, the autoregressive and cross-lagged parameters have to fall within a particular range, depending on their specific combination; if these requirements are not met, the process would be non-stationarity and diverge (see Bailey et al. (2024) for an example).

Since the data for each individual in the sample are generated according to the same time series process with the exact same parameter values, these data can be considered to reflect ergodicity. A process can be said to be ergodic if the result of taking a pooled statistic aggregated over time is equal to taking the pooled statistic aggregated over multiple participants. For this to be the case two conditions need to hold: the underlying data distributions do not change over time (stationarity) and all participants need to be so similar as for each participant to be replaceable with a randomly chosen other participant in the sample (Hunter, Fisher, & Geier, 2024). This means that not only does every individual case fluctuate around a time-invariant mean with a time-invariant variance, but that every individual case fluctuates around the *same* mean. Additionally, in an ergodic process the same dynamics hold for every individual case, and inferences made in the sample can be generalized to individuals outside the sample (Speelman, Parker, Rapley, & McGann, 2024). Especially important for our purposes is that this means there are no long-run stable mean differences between cases that could be estimated as an illusory trait.

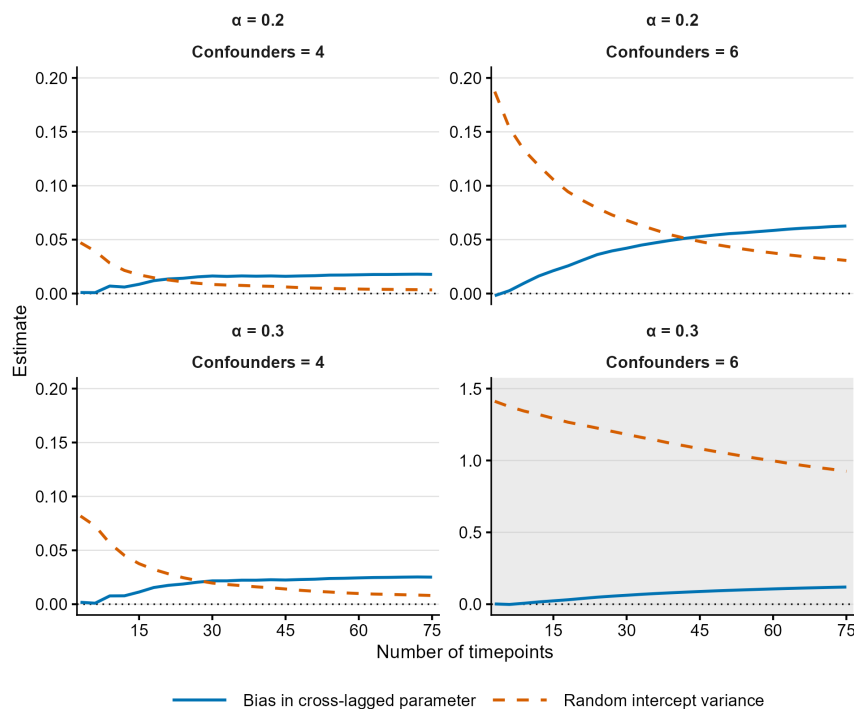
After generating 75 occasions using the model described above, we first select only the first three timepoints of the observed variables X and Y and analyze them with the CLPM and RI-CLPM, as this is the minimum number of timepoints for the latter model to be identified. Subsequently, we select the first six timepoints and fit the two models to these, and we continue this process in steps of three until we reach models estimated on all 75 timepoints of X and Y . These steps of three are chosen to limit computational cost when estimating models of this size. Parameters are constrained to be equal over time (e.g. the autoregression between X_{t1} and X_{t2} is equal to the autoregression between X_{t2} and X_{t3}) corresponding to a stationary process. All analyses are performed in R 4.3.1 (R Core Team, 2023) with models being estimated using maximum likelihood as implemented in lavaan 0.6-17 (Rosseel, 2012).

Results

Figure 2 shows the bias in the cross-lagged parameter from X to Y , together with the estimated variance of the random intercept for X , for each combination of the number of confounders and autoregressive effects. We choose to display only one of the cross-lagged parameters and one of the random intercepts since the model is symmetrical, and as such the results are virtually identical for the other parameters. As expected for all conditions, the illusory trait variance decreases as the number of timepoints increases. The bias in the cross-lagged parameter rises accordingly, illustrating the direct relationship between the illusory trait variance and bias in cross-lagged parameters. The data generating mechanism conditions show that more confounders (right column versus left column), or a more stable process due to higher autoregressive effects (bottom row versus top row) are associated with a larger variance for the illusory trait, and a slower decline of this variance as the number of timepoints increases.

Figure 2

Bias in the cross-lagged parameter and random intercept variance per condition. α = autoregressive effects of all variables (observed or unobserved), Confounders = number of unobserved confounders on both observed variables. Highlighted panel has different y-axis.



It is important to note the large increase in illusory trait variance estimated in the lower-right condition ($\alpha = .3$ combined with 6 confounders) compared to the other three conditions is due to this process being close to non-stationarity. Increasing either the number of confounders to 7 or setting the autoregressive effect to .4 would result in a process that breaks stationarity.

Concluding, as the number of timepoints increases, the ability of the RI-CLPM to correct for systems of time-varying confounding deteriorates as a result of the random intercept failing to capture this confounding variance. What is striking however, is how well the random intercept is able to capture this variance on only a small number of timepoints, and thus allowing the model to retrieve unbiased estimates of the cross-lagged parameters. This then raises the question if this property can be exploited for models concerning more timepoints.

The Segmented Random Intercept Cross-Lagged Panel Model

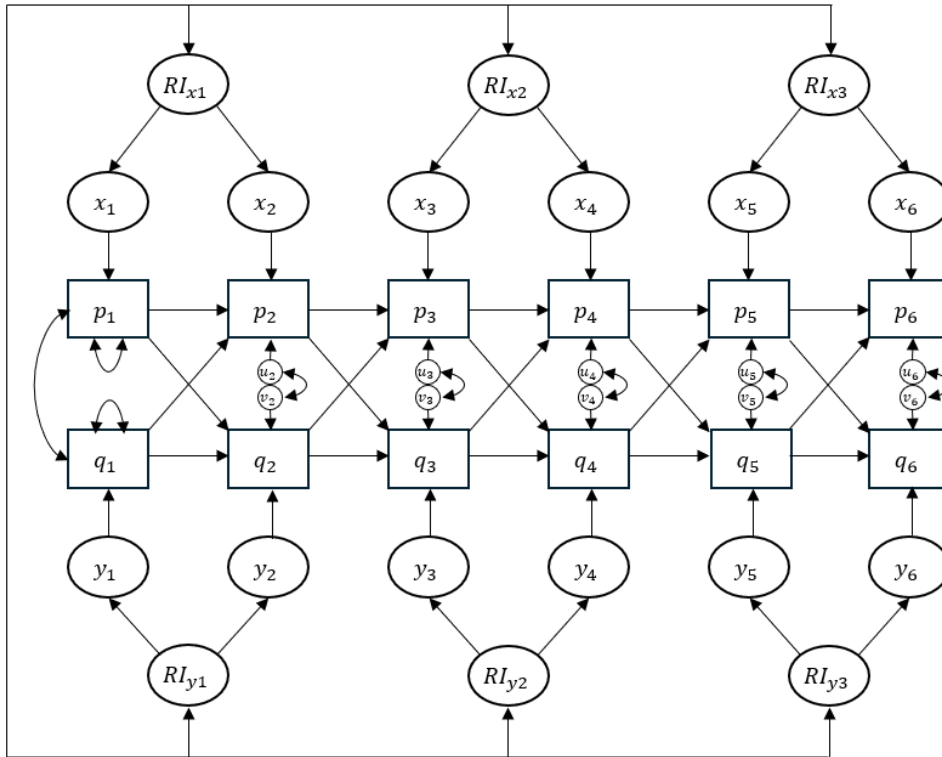
In this section we introduce a new model that aims to make use of the illusory trait finding to correct for time-varying confounding when there are more than just a few timepoints. The development of this model is based on the following reasoning: If a random intercept helps to control for time-varying confounding across a small number of timepoints, but fails to do so when the number of timepoints has increased, one approach could be to estimate multiple random intercepts on segments (subdivisions) of the total timespan. In this manner, the new model could be expected to retrieve less biased cross-lagged parameter estimates, even for data across longer timespans.

Model Specification

We propose the Segmented Random Intercept Cross-Lagged Panel Model (SRI-CLPM), which is a CLPM extended with random intercepts for segments of the total timespan of the study. Figure 3 shows a schematic example of a SRI-CLPM, specified on segments with two timepoints.

Figure 3

Schematic representation of the SRI-CLPM. RI = Random Intercept, all random intercepts are allowed to covary, represented by the connecting double arrow.



The use of the SRI-CLPM requires the user to decide on the segment length on which a segmented random intercept is specified. Making an informed decision on the segment length is more challenging than it might seem at first. A reasonable expectation is to believe segments of only two timepoints should yield the least biased estimates, since our previous results show bias increased when random intercepts are based on more timepoints. A minimal segment length of two should then work optimally, if it is identified. This presupposes however, that estimating this many latent variables does not introduce unstable estimation, especially in smaller samples. In addition to this, it is preferable to have segments of equal length, and no timepoints should be left out of a segment. This is important given that we want to be able to constrain the segmented random intercepts to denote the same underlying process, something we will explain further in the next paragraph. Keeping the segments of equal length essentially means the total number of

timepoints should be divisible by the segment length, once again making the minimum of two an attractive first choice. We further investigate the effect of segment length with the simulation presented below.

Another concern is how to estimate the covariance structure of the latent variables. Imagine a bivariate SRI-CLPM with 30 measurements and random intercepts estimated on a segment length of two. This will result in 30 separate random intercepts (15 for each repeatedly measured variable), each with a variance that needs to be estimated. The number of possible unique covariances between these random intercepts is 465. Estimating the total latent variable covariance matrix freely thus results in 495 parameters being estimated for the latent variables alone. This is much more than the 3 parameters (two variances and a covariance) that would be included in the standard RI-CLPM for the two overall random intercepts; it could pose a problem concerning model parsimony in the best case, and empirical identifiability in the worst case. Therefore, we choose to impose restrictions on the variances and covariances between latent variables.

A natural starting point here is to impose constraints that follow the stationarity assumption along which we also simulate the data. This implies we constrain the variances of all segmented random intercepts on the same variable to be equal, since we assume the variance in the confounding process to be stable over time. In addition, we also constrain covariances across a set lag to be equal. In our bivariate example, this would imply two distinct sets of constraints; covariances between ‘lateral’ random intercepts on the same variable (i.e., two random intercepts on X) and covariances between random intercepts on separate variables (i.e., a random intercept on X and one on Y). Both sets of covariances are constrained to be equal across a set lag, but both are constrained separately. For example, the covariance between the first and second random intercepts on X are constrained equal to the covariance between the second and third. The same holds for covariances between a random intercept on X and one on Y , but these are not constrained to the same covariance at the same lag on two random intercepts estimated on the same variable. Constraining the random intercept covariances in this manner corresponds to a block-Toeplitz

matrix, which can be seen as minimally restrictive (assuming stationarity) while still reducing the number of parameters greatly. A schematic representation of the resulting constrained covariance matrix between the segmented random intercepts can be found below, alongside the formula for the number of degrees of freedom of a SRI-CLPM constrained in this manner.

Table 1

Constrained segmented random intercept covariance matrix, shading to illustrate the different ‘blocks’ corresponding to distinct kinds of covariances.

	RIx1	RIx2	RIx3	RIx4	RIy1	RIy2	RIy3	RIy4
RIx1	$\psi_{x,x}$							
RIx2	$\psi_{x,x}^{lag1}$	$\psi_{x,x}$						
RIx3	$\psi_{x,x}^{lag2}$	$\psi_{x,x}^{lag1}$	$\psi_{x,x}$					
RIx4	$\psi_{x,x}^{lag3}$	$\psi_{x,x}^{lag2}$	$\psi_{x,x}^{lag1}$	$\psi_{x,x}$				
RIy1	$\psi_{x,y}^{lag0}$	$\psi_{x,y}^{lag1}$	$\psi_{x,y}^{lag2}$	$\psi_{x,y}^{lag3}$	$\psi_{y,y}$			
RIy2	$\psi_{x,y}^{lag1}$	$\psi_{x,y}^{lag0}$	$\psi_{x,y}^{lag1}$	$\psi_{x,y}^{lag2}$	$\psi_{y,y}^{lag1}$	$\psi_{y,y}$		
RIy3	$\psi_{x,y}^{lag2}$	$\psi_{x,y}^{lag1}$	$\psi_{x,y}^{lag0}$	$\psi_{x,y}^{lag1}$	$\psi_{y,y}^{lag2}$	$\psi_{y,y}^{lag1}$	$\psi_{y,y}$	
RIy4	$\psi_{x,y}^{lag3}$	$\psi_{x,y}^{lag2}$	$\psi_{x,y}^{lag1}$	$\psi_{x,y}^{lag0}$	$\psi_{y,y}^{lag3}$	$\psi_{y,y}^{lag2}$	$\psi_{y,y}^{lag1}$	$\psi_{y,y}$

The degrees of freedom for a Toeplitz constrained SRI-CLPM can then be expressed as:

$$DF = T(2T + 1) - (9 + 4(\frac{T}{SL} - 1) + 2T)$$

Where T denotes the number of timepoints and SL denotes the segment length.

Comparing this to the expression for the degrees of freedom of the unconstrained latent variable covariances below shows the utility of these constraints.

$$DF = T(2T + 1) - (9 + 2(\frac{T}{SL})^2 + 2T)$$

Investigating the Performance of the SRI-CLPM

Now that we have specified the SRI-CLPM, we evaluate its performance and behavior with a series of simulation studies. The first is a larger study aimed at investigating the degree of bias in the cross-lagged parameters as more timepoints are considered. The second is a more focused, exploratory study aimed at investigating the statistical power and type I error rates for cross-lagged parameters as estimated in the SRI-CLPM.

Bias in the Cross-Lagged Parameters

In order to evaluate the degree of bias in the cross-lagged parameters retrieved by the SRI-CLPM, we perform a Monte Carlo simulation. Once again, we generate data according to the data generating mechanism described in Bailey et al. (2024), corresponding to a CLPM with systems of time-varying confounding. Similar to the first simulation, we choose to estimate models on differing numbers of timepoints to evaluate the performance of the model as more timepoints are considered. But in contrast to the first simulations, we make use of smaller, more realistic sample sizes (i.e., 250 and 500 cases) to assess the practical utility for researchers. We make use of 3000 replications in each condition to study sampling variation, allowing us to retrieve empirical sampling distributions for the model parameters.

The study length that we consider was chosen to be 6, 12, 18, 24 and 30 timepoints. This is done so all numbers of timepoints can be divided by both 2 and 3, allowing us to specify segments of these lengths. In this manner we aim to get some indication of the role of segment length in the performance of the model. To assess the performance of the Toeplitz-constrained random intercept covariances we estimate two specifications of the SRI-CLPM: one with these constraints and one with freely estimated random intercept variances and covariances.

Figure 4 shows the distributions of one of the cross-lagged parameter estimates, visualized as boxplots, as they are obtained from various models and for an increasing number of timepoints. The true parameter value is .1, displayed as the horizontal dotted line. The RI-CLPM's cross-lagged parameter estimates slowly increase to the uncorrected biased estimates retrieved by the CLPM. As expected, the SRI-CLPM is able to retrieve less biased cross-lagged parameter

estimates when the number of timepoints considered increases, at the cost of somewhat larger variance in the sampling distribution compared to the CLPM and RI-CLPM. Figure 4 also shows the effect of the segment length chosen for the random intercepts has on the bias and sampling variability of the cross-lagged parameters. Increasing the segment length from two to three leads to a slight increase in bias, but in turn also reduces sampling variability slightly, which is likely due to the decrease in the number of parameters estimated that needs to be estimated when the segment is larger.

Figure 4

Sample distributions of the cross-lagged parameter per model specification. Boxplots represent the interquartile range of point estimates. Dashed line represents the true value of .1. SL = segment length, -free indicates the model had freely estimated covariances between segmented random intercepts.

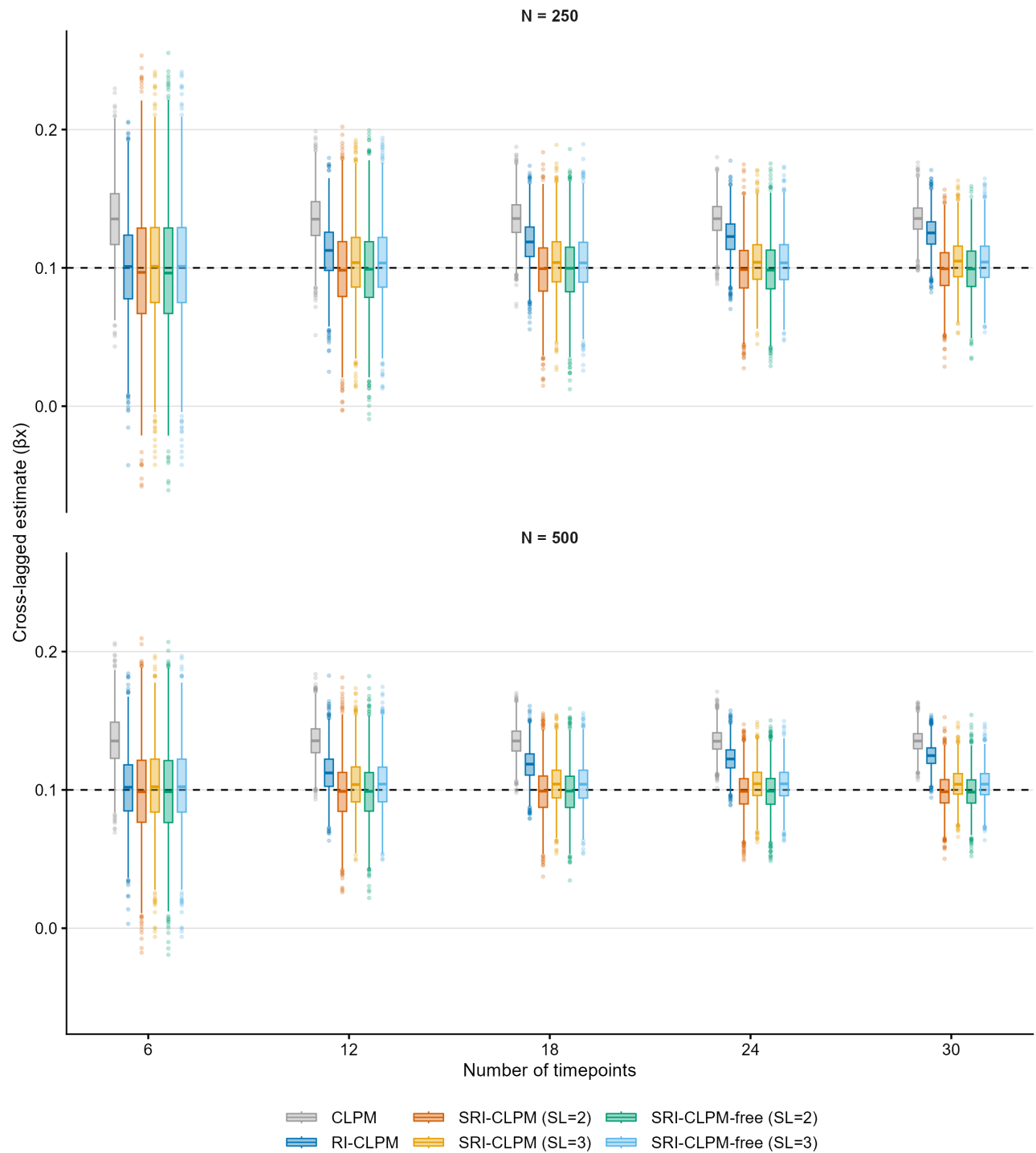
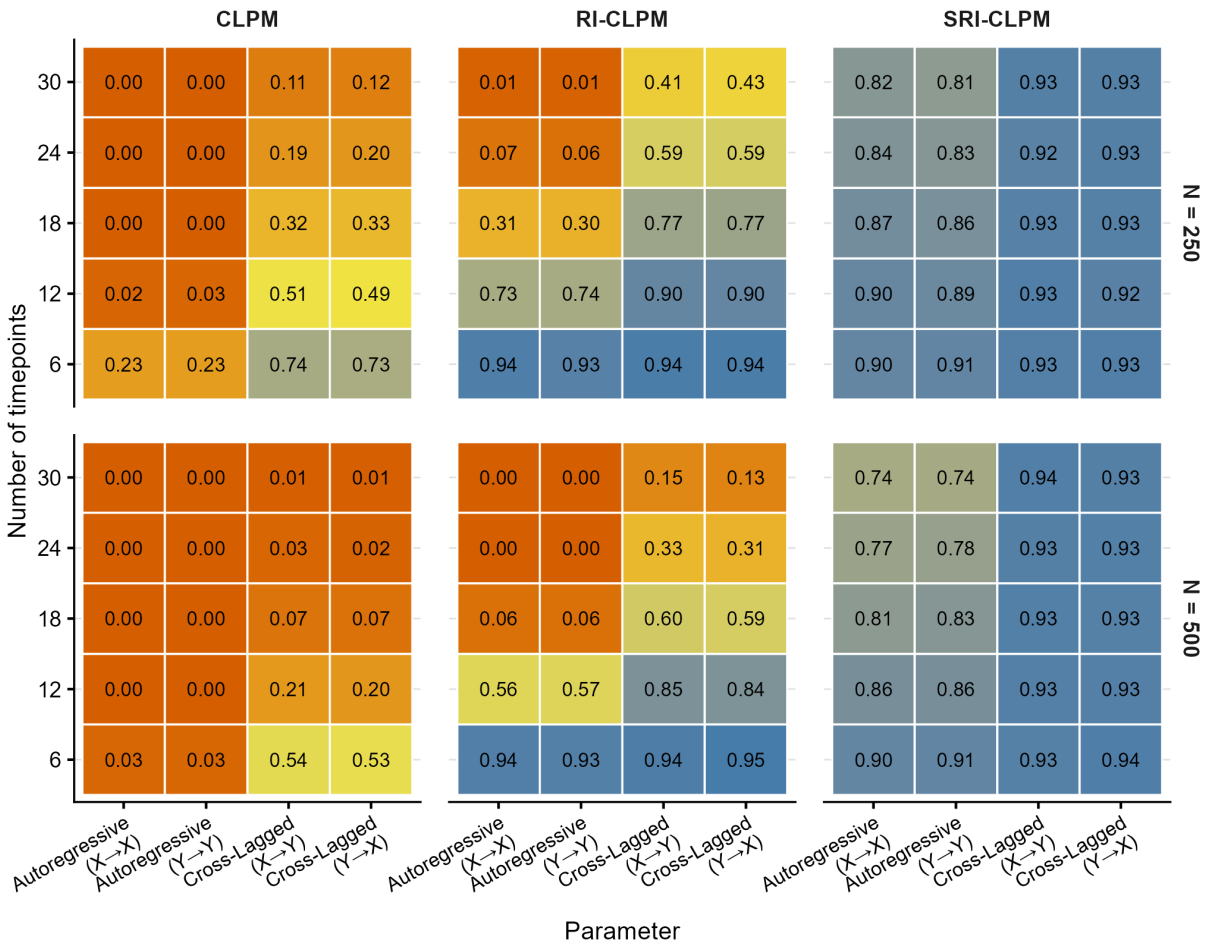


Figure 5 shows the 95% confidence interval coverage rates for all structural parameters (also including the autoregressive parameters), comparing the SRI-CLPM (right column) with Toeplitz latent variable constraints and a segment length of 2 to the CLPM (left column) and RI-CLPM (middle column). The coverage rates for cross-lagged parameters show a substantial improvement over the biased estimates in the RI-CLPM and CLPM. Interestingly however, the autoregressive parameters show a marked decline in 95% CI coverage rates as more timepoints are considered. Another puzzling aspect of this is that it seems the model decline is more pronounced on the larger sample size, which is somewhat alarming. It is important to note however, that retrieving unbiased estimates of autoregressive parameters was not the goal of this simulation and is usually not the goal of cross-lagged panel modeling.

Figure 5

Heatmap showing the 95% confidence interval coverage rates for all structural parameters per model, per timepoint and per sample size condition (N). Coverage rate denotes the frequency with which the true value was contained in the 95% confidence interval.



Investigating Statistical Power and Type I Error with the SRI-CLPM

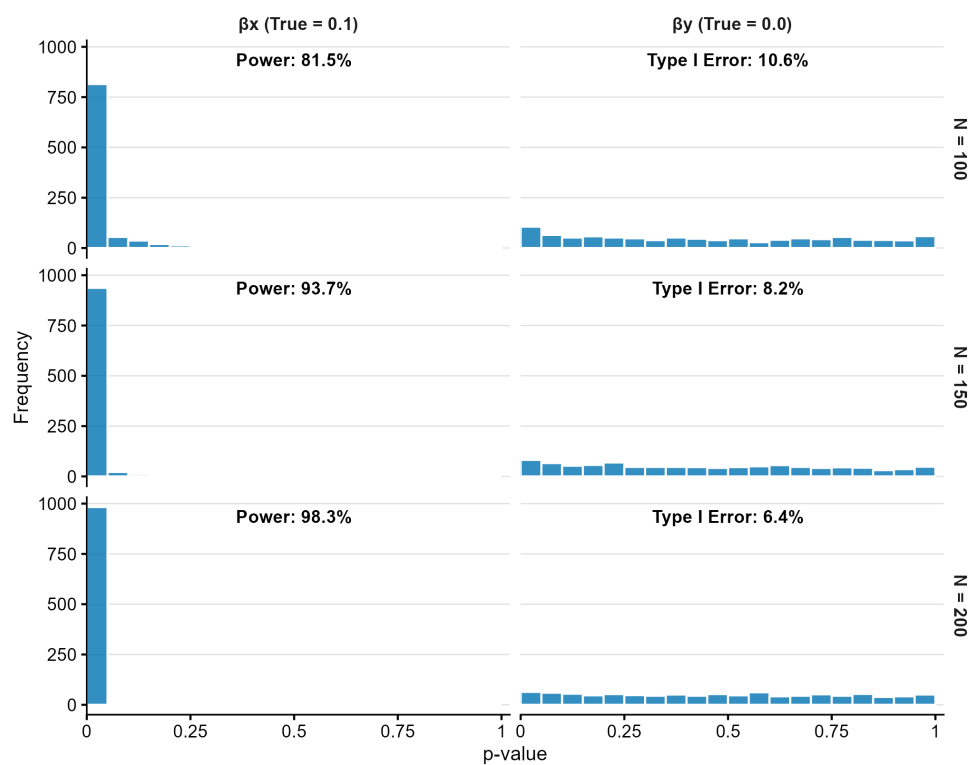
We present an additional simulation study aimed at investigating the statistical power and type I error rates for the cross-lagged parameters as estimated in the SRI-CLPM. To achieve this, we generate data from the same mechanism as before, only setting one of the cross-lagged effects to zero. This allows us to assess both statistical power in detecting the true cross-lagged effect of .1, and the type I error rate of detecting an effect when the true cross-lagged effect is zero. For simplicity, we focus on 20 timepoints, using a segment length of two, while varying the sample sizes between 100, 150 and 200. For each of these three conditions, we simulated a thousand

replications.

The distribution of p-values for the cross-lagged parameters are presented in Figure 6 below, along with the statistical power and type I error. The SRI-CLPM shows acceptable power on samples as small as 100 cases and a relatively uniform p-value distribution on the cross-lagged effect that was set to zero. There is however a somewhat inflated type I error rate when compared to the nominal α of .05 used, which importantly does decrease on larger sample sizes.

Figure 6

Distributions of estimated p-values per cross-lagged parameter true value (0.1 or 0) per sample size condition (N). Bar chart shows the distributions, power denotes the percentage where the null hypothesis was correctly rejected, type I error denotes the percentage where the null hypothesis was incorrectly rejected.



It is important to note that during the estimation of the SRI-CLPM's the covariance matrix of latent variables is often not positive definite. Additionally, lavaan also incidentally reports the variance-covariance matrix of the estimated parameters to not be positive definite. On inspection of an estimated correlation matrix for a SRI-CLPM with freely estimated latent variable

covariances, we found that adjacent segmented random intercepts are structurally estimated to be correlated higher than 1. This reflects an improper solution found by the estimator; possibly indicating a structural problem with empirical identifiability. This problem persists even if models are fitted on very large amounts of data (50.000 cases) and using the Toeplitz-constraints. For now we have reported the estimated cross-lagged parameters, since these do not seem to be anomalous, and will comment more on estimation in the discussion.

Concluding, the SRI-CLPM is able to retrieve less biased estimates of the cross-lagged parameters when the number of timepoints increases compared to the RI-CLPM and CLPM. This comes at the cost of somewhat larger standard errors, yet acceptable statistical power in the condition we simulated. The Toeplitz-constrained covariances between the random intercepts work well for the stationary, ergodic process used, with the choice of segment length

The SGRI-CLPM

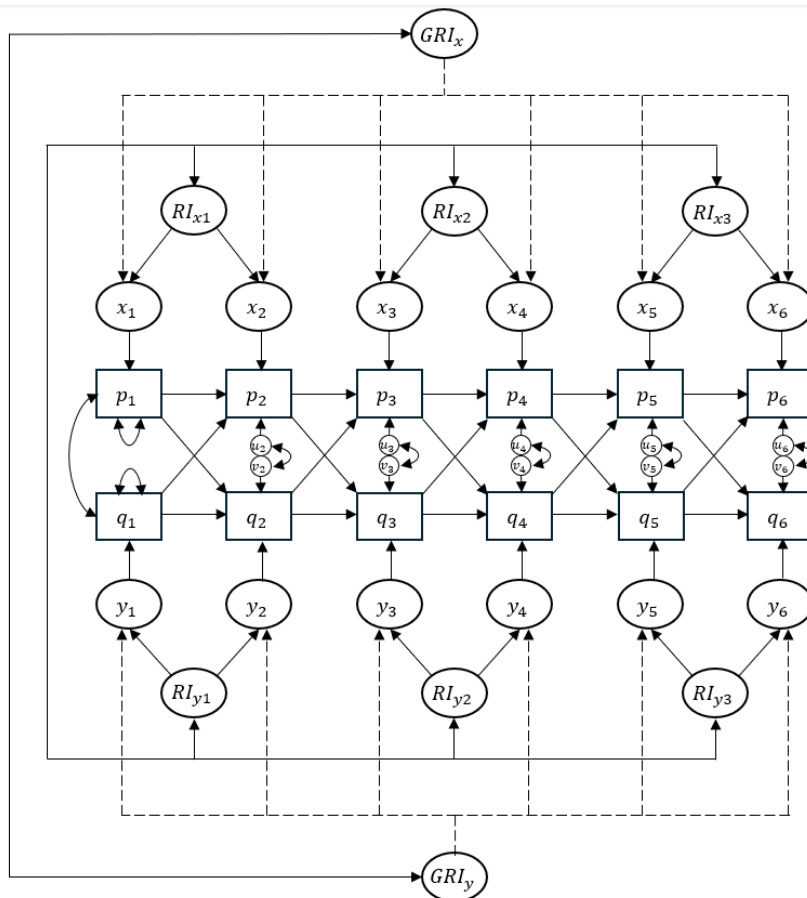
Until now, we have only considered processes containing systems of time-varying confounding, introducing the SRI-CLPM to control for its influence. We therefore have the RI-CLPM, designed to control for time-invariant confounding and the SRI-CLPM, designed for controlling for time-varying confounding. This raises the question however, to what extent the SRI-CLPM is able to control for time-invariant confounding or a combination of both types of confounding. Additionally, it might be interesting to see whether it is possible to separate the confounding variance into a time-varying and time-invariant component, allowing researchers to get an idea of the source of the confounding. We therefore in this section introduce the idea of adding a ‘global’ random intercept to the SRI-CLPM, aimed at capturing time-invariant confounding. The motivation for this is simple: We expect there to be substantial time-invariant confounding in any process of interest, especially concerning person-to-person differences in the social sciences. Therefore, a model which could correct for the influence of both types of confounding while accurately partitioning their associated variance would be an asset.

Model Specification

Aiming to capture time-invariant confounding, we specify a variation of the SRI-CLPM with the addition of a global random intercept estimated on all repeated measurements of a variable, resembling the standard RI-CLPM. These global random intercepts can be allowed to covary freely, but are assumed to be independent of the other segmented random intercepts. In this manner, we hope to be able to distinguish variance associated with time-invariant confounding from variance associated with time-varying confounding. A schematic representation of this new model, which we refer to as the SGRI-CLPM, is shown in Figure 7 below.

Figure 7

Schematic representation of the SGRI-CLPM. RI = random intercept, GRI = global random intercept. Random intercepts are allowed to covary, but their covariance with the global random intercepts is constrained to zero.



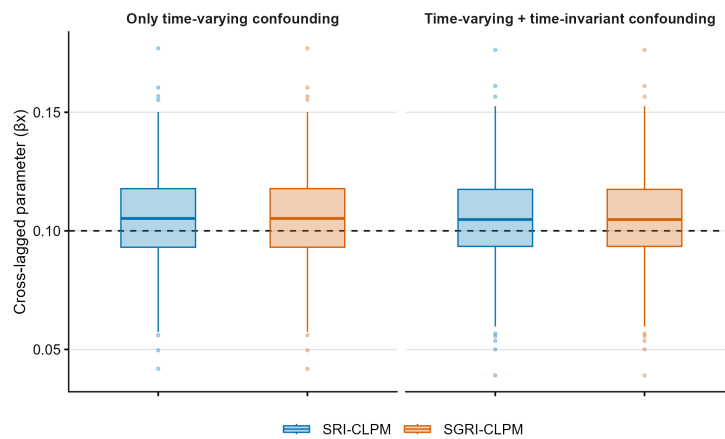
Investigating Cross-Lagged Parameter Bias in the SGRI-CLPM

We assess the performance of the SGRI-CLPM with another Monte Carlo simulation study, comparing it to the performance of the SRI-CLPM. For this, the data generating mechanism introduced by Bailey et al. (2024) is extended to also allow for the generation of data with time-invariant confounding. We assume the time-invariant confounding is orthogonal to the time-varying confounding, meaning we only have to generate a between-case deviation from the mean. This entails specifying a variance for both the confounder on X and on Y , taking into account their covariance. Afterwards, this deviation from the mean per case can be added to its time-varying process.

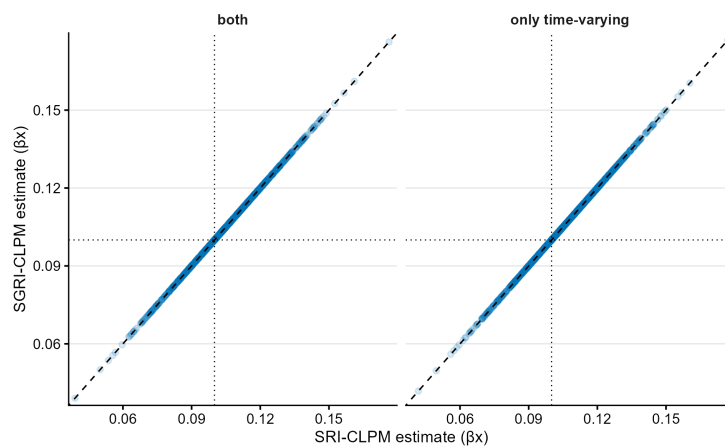
Both the SRI-CLPM and this extended SGRI-CLPM are then used to analyze the data generating with various mechanisms: either with only time-varying confounding or with both types of confounding. Both specifications use the Toeplitz-constrained segmented random intercept covariances and a segment length of three, on 30 total timepoints. While earlier simulations show a smaller segment length yielding better results, three was chosen as a convenient way to divide 30 timepoints. We simulate 1000 replications, using a realistic sample size of 200. The distributions of cross-lagged parameter estimates for the two model in each of the conditions are plotted in Figure 8. Interestingly, the model specifications seem to perform identically at retrieving unbiased cross-lagged parameter estimates, even when a time-invariant component is added. This becomes especially clear in Figure 9, where the point estimate of the SGRI-CLPM is plotted against the point estimate of the SRI-CLPM on the same data, which shows that these are virtually identical.

Figure 8

Interquartile ranges for estimated cross-lagged parameters per model, per confounding condition. Dashed line represents the true value of .1.

**Figure 9**

Point estimates for both model specifications on the same data plotted against one another, per confounding condition. Dashed line represents equality between both estimates.



Hence, the only potential added benefit of a global random intercept may be its alleged ability to partition confounding variance into a time-varying and time-invariant component. To assess this ability we should look at the estimated global random intercept variances and their associated covariance.

Random Intercept Variances in the SGRI-CLPM

The SGRI-CLPM estimates two random intercept variances per variable of interest, assuming all the segmented random intercept variances are constrained to be equal. We estimate a global random intercept variance representing a stable, time-invariant confounding component and a segmented random intercept variance representing fleeting, time-varying confounding. We have already seen that adding the segmented random intercepts is sufficient for correcting for the influence of both time-varying and time-invariant confounding on the estimation of cross-lagged parameters. However, researchers might also be interested in uncovering the source of the confounding, to which end separating the time-varying and time-invariant components could be useful.

To assess how accurately the SGRI-CLPM can separate the confounding variance into the time-varying and time-invariant components, we generate data with all four combinations of confounding. This means we generate data according to a simple CLPM without confounding, a CLPM with time-varying confounding, a CLPM with time-invariant confounding (i.e., RI-CLPM) and a CLPM with both types of confounding. We can then compare the estimated segmented random intercept variance and the global random intercept variance across these conditions. All conditions are replicated 500 times, with a sample size of 200 and using a SGRI-CLPM estimated on 20 timepoints with a segment length of two. It should be noted at this point that the values for the segmented random intercept and global random intercept cannot be directly compared. The global random intercept variance is attributed to one underlying latent variable, indicated by all measurements and should therefore in theory retrieve the variance specified for the time-invariant component in the data generating mechanism. In contrast, the segmented random intercept variance is constrained to be equal across multiple latent variables, each having an effect on a segment of the total timespan. The estimated segmented random intercept variance therefore does not necessarily retrieve one inputted variance in the data generating mechanism.

Table 2 shows the mean random intercept variances of X per condition, alongside the mean point estimate of the cross-lagged parameter of X on Y . When the unconfounded CLPM

generated the data, the SGRI-CLPM correctly estimate random intercept variances around 0 for both the segmented and global random intercepts, but it does suffer from improper solutions by estimating negative variances. Interestingly, the global random intercept seems to be able to capture the stable component relatively well. In the conditions where a time-invariant confounder was present (time-invariant and combined) the mean estimate is relatively close to the true value of 1. There does seem to be some misattribution of variance into the other components, which is most clearly visible in the time-invariant condition. We would expect the global random intercept to capture the complete confounding variance of 1, yet a small portion of the variance is attributed to the segmented random intercepts. The same holds for the time-varying condition, although the difference here is harder to interpret because we have not quantified the total time-varying component. The combined condition shows a promising separation, although the time-varying component seems to be somewhat inflated.

Table 2

Mean of Point Estimates for SGRI Model Parameters Across Simulation Conditions

Condition	Global RI	Segmented RI	Cross-Lagged Parameter
Pure (No Confounding)	-0.008	0.008	0.102
Time-varying	0.034	0.041	0.100
Time-invariant	0.968	0.026	0.101
Combined	1.010	0.059	0.099

These results should be seen as exploratory, and a proof of concept. Further work is needed to validate the SGRI-CLPM's ability to separate confounding into its source components. We only show here that it is in theory possible to make some sort of separation, whether this separation can be made cleanly and accurately cannot be said from this simulation alone. If a researchers only interest is to control for the influence of unobserved confounding, whether time-varying or time-invariant, the SRI-CLPM suffices.

Discussion

In this paper, we aim to apply the ‘illusory trait’ finding as presented by Bailey et al. (2024) to control for time-varying confounding in cross-lagged panel modeling when the number of timepoints increases. Unobserved confounding is a serious threat to the estimation of structural parameters in SEM (Hamaker et al., 2015; Murayama & Gfrörer, 2024; Rogosa, 1980). While theoretically identifying possible confounders and controlling for their influence through their modeling as covariates may be considered preferable, it is unlikely that we are able to always observe all potential confounders. We therefore introduced a new model we refer to as the Segmented Random Intercept Cross-Lagged Panel Model (SRI-CLPM), which consists of extending the CLPM with random intercepts for segments of the total study duration. Simulation results show the SRI-CLPM successfully controls for confounding, whether time-varying, time-invariant, or both. We believe a model that is able to retrieve unbiased cross-lagged parameters regardless of the presence or absence of any type of confounding to be of substantive interest to researchers.

In the first simulation, we have shown that the illusory random intercept variance decreases as the number of timepoints increases. As a result, the ability of the RI-CLPM to retrieve less biased estimates of the cross-lagged parameters deteriorates until it eventually retrieves the uncorrected bias as in the CLPM. Given the random intercepts apparent ability to correct for time-varying confounding in the short term, we then utilized this by designing a model with multiple random intercepts estimated on segments of the full timespan. We then assess the performance of this SRI-CLPM in the second simulation. Here we have shown the SRI-CLPM is able to retrieve substantially less biased estimates of the cross-lagged parameters on longer timespans at the cost of larger standard errors. Interestingly, the same did hold for the autoregressive parameters, yet as more timepoints are considered this unbiased estimation seems to deteriorate. We have then proposed another possible modeling specification which makes use of a global random intercept, specified in the same way as in the traditional RI-CLPM, with its covariance to the segmented random intercepts fixed to zero. This was aimed at separating

time-invariant confounding variance from time-varying confounding. The performance of this model was compared to the simpler SRI-CLPM on datasets containing both time-varying and time-invariant confounding. Both model specifications retrieved almost exactly the same, unbiased cross-lagged parameter estimates. Although the addition of the global random intercept shows some promise in being able to separate confounding into its sources, more work is needed to validate these results.

While these initial findings are promising, there remains considerable uncertainty about the practical functionality of the SRI-CLPM and its extension, the SGRI-CLPM. Retrieving unbiased estimates of the cross-lagged parameters is certainly a very desirable trait for an extension of the CLPM. However, other potentially important aspects, such as bias in the autoregressive parameters or inference using the random intercepts, have largely been ignored in the present study. The slow increase in bias in the autoregressive parameters as we consider more timepoints is particularly interesting, even though we expect this to be of limited importance to researchers. While inferences could be made about stability in the variables of interest along the autoregressive parameters, these parameters are mostly intended to fulfill a controlling role for the estimation of cross-lagged parameters. It is also important to underline that the SRI-CLPM still retrieved more accurate estimates than the other models. Nevertheless, much is still unclear about the precise conditions which are needed for the SRI-CLPM to perform better than its alternatives. We therefore advise researchers to exercise caution in using the S(G)RI-CLPM as a model of first choice for panel data or ILD without considering whether their study design necessitates its use. The much more parsimonious and well-studied RI-CLPM already provides substantial correction against both types of confounding, as shown by the illusory trait finding, and should therefore be preferred in most scenarios until further research confirm the utility of the SRI-CLPM. If researchers are analyzing data in which the presence of substantial confounding is assumed, or a large number of timepoints are measured, estimating the SRI-CLPM could be considered. Additionally, researchers could utilize the SRI-CLPM as a sort of sensitivity check, providing some insight into the possibility of time-varying confounding. If researchers find their

cross-lagged parameter estimates substantially altered after fitting a SRI-CLPM compared to a RI-CLPM, this might be an indication that unobserved time-varying confounding cannot be ruled out.

Furthermore, it will be of interest to compare and contrast the SRI-CLPM with the LTVC-CLPM introduced in Kenny and McCoach (2025). At first glance these models may seem similar, but there are some important distinctions in how time-varying confounding is addressed. Both models make use of latent variables to control for time-varying confounding, but differ in their implementation. In the LTVC-CLPM, time-varying confounding is modeled as a series of latent variables indicated by both variables in a wave, with subsequent latent variables being regressed on the previous in an autoregressive process. This is in contrast to the SRI-CLPM, where the latent variables are indicated by a segment of multiple measurements of the same variable, with the segmented random intercepts being allowed to covary. We believe these differences in latent variable modeling explain the difficulties in estimating the LTVC-CLPM compared to the SRI-CLPM. In the LTVC-CLPM measurements of a variable are connected to previous measurements not only through the autoregressive parameters, but also through a web of complex interdependencies caused by the regressive paths through the latent confounders. The SRI-CLPM avoids this issue by covarying the latent variables instead of regressing them, thus simplifying the relations in the structural model. While in the original paper the LTVC-CLPM could not be successfully estimated, a more recent paper by Yu and Finkel (2026) used a similar approach to correct for time-varying confounding in a longitudinal fixed effects regression model. This specification was successful in reducing unobserved confounding bias, underlining the possible utility of this approach. More research should therefore be done to investigate other applications of the LTVC approach to different models.

There are several important limitations to consider when interpreting the current study. First, it is important to note that the current simulations are based on a data generating mechanism which was designed to simulate panel data, which does not necessarily resemble ILD. The data generating mechanism was adapted from Bailey et al. (2024), who informed their choice of

parameter values on average panel data values reported in the literature as compiled by Orth et al. (2021). In reality, ILD parameters might differ substantially from these reported panel data values, something the current study did not consider. Second, an underexplored aspect of the current study is the estimation of the model, including the prevalence of negative latent variable variances estimates. For most estimated SRI-CLPMs lavaan reports a negative latent variable variance, with an additional warning given at times for a non-positive definite covariance matrix of latent variables. We have chosen to ignore these warnings in this paper since the models seem to retrieve correct values for parameters of theoretical interest. It should be recognized however, that this needs to be investigated further, especially since ignoring warnings should not be made common practice for researchers. Third, in the current study we have not assessed overall model fit. Therefore, it is unknown whether a researcher would end up correctly or incorrectly choosing the SRI-CLPM over the RI-CLPM when selecting a model using a likelihood-ratio test or fit measures. This is of additional importance given the finding by Bailey and colleagues (2024) that one cannot rely solely on model fit in deciding whether a significant random intercept variance implies a stable trait, since it could in fact be illusory. Fourth, the current model specification is constrained to be rather restrictive. We have only considered processes that conform to both strict stationarity and ergodicity. While we believe the stationarity assumption is a realistic and helpful constraint in many settings and especially in psychology, it is not appropriate for all theoretical processes or study designs (e.g. pre-post designs). Similarly, assuming ergodicity might not always be desirable or defensible in all processes, although it is perhaps already assumed too often (see for example: Molenaar, 2004; Speelman et al., 2024).

In sum, there are many aspects, both theoretical and practical, of the new model that are still unclear. Future work should therefore be done to clarify the functionality of the model, under which conditions it retrieves accurate estimates and how it can be extended. For example, in the current study lavaan often retrieves improper solutions concerning the latent variables, such as correlations of more than 1 and negative random intercept (co)variances. Future work could investigate whether these issues could be addressed by using different estimators, through

constrained maximum likelihood or Bayesian estimation with constrained priors. Another important area of study is model selection and model fit, by comparing the S(G)RI-CLPM with the more parsimonious RI-CLPM and CLPM. Finding a reliable manner of model selection between these four possible cross-lagged panel models could be a valuable asset, since we have now introduced a model for every possible combination of time-varying and time-invariant confounding. Additionally, work could be done to relax the restrictions currently in place by assuming stationarity and ergodicity. An interesting extension could be made to allow for individual cases to have random slopes for the cross-lagged effects, thereby relaxing the assumption of ergodicity.

Concluding, we want to emphasize the importance of considering the possible and very likely presence of confounding in any process of interest, and how to control for its influence. As the present study and the original paper by Bailey et al. (2024) shows, the RI-CLPM is able to correct for both time-invariant confounding and some influence of time-varying confounding, yet this correction for time-varying confounding breaks down as the number of timepoints measured increases. Therefore, researchers have the option of exploring the newly introduced S(G)RI-CLPM, which is shown to be able to retrieve less biased structural parameter estimates compared to the RI-CLPM. While much is still unclear about this model, it provides a promising and reasonably accessible method for addressing confounding, regardless of its source.

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